

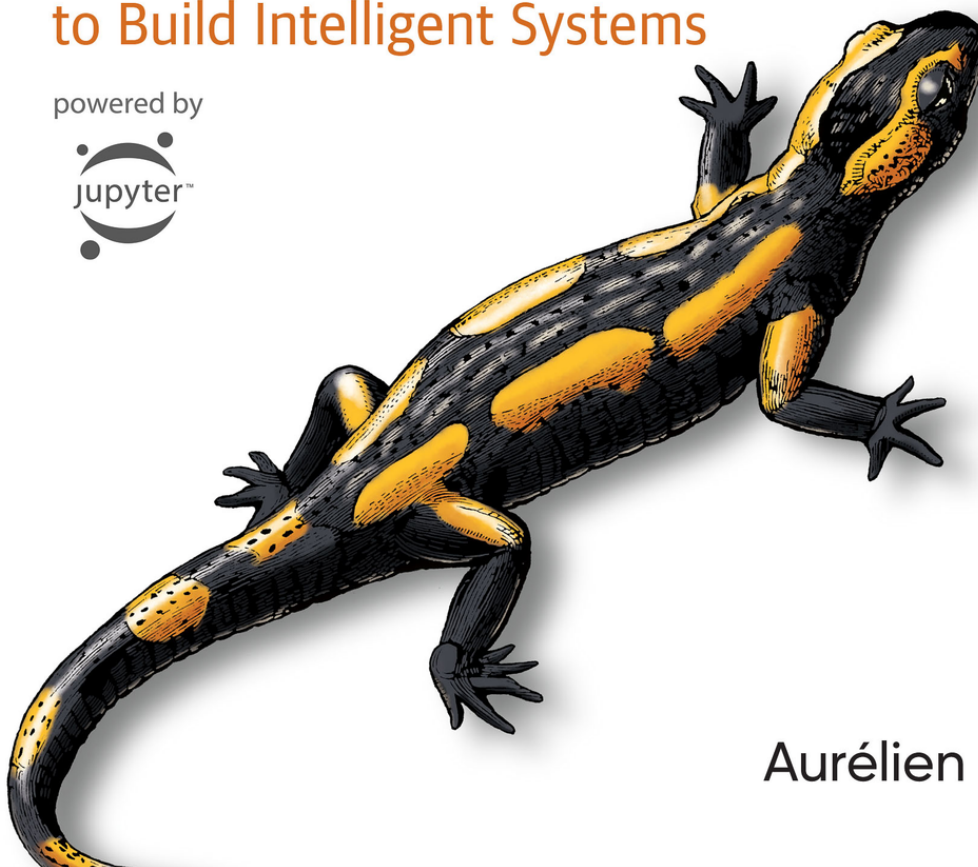
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Third
Edition

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
to Build Intelligent Systems

powered by



Aurélien Géron

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

THIRD EDITION

Concepts, Tools, and Techniques to Build Intelligent Systems

Aurélien Geron

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

by Aurélien Geron

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Preface

The Machine Learning Tsunami

In 2006, Geoffrey Hinton et al. published a [paper](#)¹ showing how to train a deep neural network capable of recognizing handwritten digits with state-of-the-art precision (>98%). They branded this technique “deep learning”. A deep neural network is a (very) simplified model of our cerebral cortex, composed of a stack of layers of artificial neurons. Training a deep neural net was widely considered impossible at the time,² and most researchers had abandoned the idea in the late 1990s. This paper revived the interest of the scientific community, and before long many new papers demonstrated that deep learning was not only possible, but capable of mind-blowing achievements that no other machine learning (ML) technique could hope to match (with the help of tremendous computing power and great amounts of data). This enthusiasm soon extended to many other areas of machine learning.

A decade later, machine learning had conquered the industry, and today it is at the heart of much of the magic in high-tech products, ranking your web search results, powering your smartphone's speech recognition, recommending videos, and perhaps even driving your car.

Machine Learning in Your Projects

So, naturally you are excited about machine learning and would love to join the party!

Perhaps you would like to give your homemade robot a brain of its own? Make it recognize faces? Or learn to walk around?

Or maybe your company has tons of data (user logs, financial data, production data, machine sensor data, hotline stats, HR reports, etc.), and more than likely you could unearth some hidden gems if you just knew where to look. With machine learning, you could accomplish the following **and much more**:

- Segment customers and find the best marketing strategy for each group.
- Recommend products for each client based on what similar clients bought.
- Detect which transactions are likely to be fraudulent.
- Forecast next year's revenue.

Whatever the reason, you have decided to learn machine learning and implement it in your projects. Great idea!

Objective and Approach

This book assumes that you know close to nothing about machine learning. Its goal is to give you the concepts, tools, and intuition you need to implement programs capable of *learning from data*.

We will cover a large number of techniques, from the simplest and most commonly used (such as linear regression) to some of the deep learning techniques that regularly win competitions. For this, we will be using production-ready Python frameworks:

- **Scikit-Learn** is very easy to use, yet it implements many machine learning algorithms efficiently, so it makes for a great entry point to learning machine learning. It was created by David Cournapeau in 2007, and is now led by a team of researchers at the French Institute for Research in Computer Science and Automation (Inria).
- **TensorFlow** is a more complex library for distributed numerical computation. It makes it possible to train and run very large neural networks efficiently by distributing the computations across potentially hundreds of multi-GPU (graphics processing unit) servers. TensorFlow (TF) was created at Google and supports many of its large-scale machine learning

applications. It was open sourced in November 2015, and version 2.0 was released in September 2019.

- **Keras** is a high-level deep learning API that makes it very simple to train and run neural networks. Keras comes bundled with TensorFlow, and it relies on TensorFlow for all the intensive computations.

The book favors a hands-on approach, growing an intuitive understanding of machine learning through concrete working examples and just a little bit of theory.

TIP

While you can read this book without picking up your laptop, I highly recommend you experiment with the code examples.

Code Examples

All the code examples in this book are open source and available online at <https://github.com/ageron/handson-ml3>, as Jupyter notebooks. These are interactive documents containing text, images, and executable code snippets (Python in our case). The easiest and quickest way to get started is to run these notebooks using Google Colab: this is a free service that allows you to run any Jupyter notebook directly online, without having to install anything on your machine. All you need is a web browser and a Google account.

NOTE

In this book, I will assume that you are using Google Colab, but I have also tested the notebooks on other online platforms such as Kaggle and Binder, so you can use those if you prefer. Alternatively, you can install the required libraries and tools (or the Docker image for this book) and run the notebooks directly on your own machine. See the instructions at <https://homl.info/install>.

This book is here to help you get your job done. If you wish to use additional content beyond the code examples, and that use falls outside the scope of fair use guidelines, (such as selling or distributing content from O'Reilly books, or incorporating a significant amount of material from this book into your product's documentation), please reach out to us for permission, at permissions@oreilly.com.

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Prerequisites

This book assumes that you have some Python programming experience. If you don't know Python yet, <https://learnpython.org> is a great place to start. The official tutorial on [Python.org](https://python.org) is also quite good.

This book also assumes that you are familiar with Python's main scientific libraries—in particular, [NumPy](#), [Pandas](#), and [Matplotlib](#). If you have never used these libraries, don't worry; they're easy to learn, and I've created a tutorial for each of them. You can access them online at <https://homl.info/tutorials>.

Moreover, if you want to fully understand how the machine learning algorithms work (not just how to use them), then you should have at least a basic understanding of a few math concepts, especially linear algebra. Specifically, you should know what vectors and matrices are, and how to perform some simple operations like adding vectors, or transposing and multiplying matrices. If you need a quick introduction to linear algebra (it's really not rocket science!), I provide a tutorial at <https://homl.info/tutorials>. You will also find a tutorial on differential calculus, which may be helpful to understand how neural networks are trained, but it's not entirely essential to grasp the important concepts. This book also uses other mathematical concepts occasionally, such as exponentials and logarithms, a bit of probability theory, and some basic statistics concepts, but nothing too advanced. If you need help on any of these, please check out <https://khanacademy.org>, which offers many excellent and free math courses online.

Roadmap

This book is organized in two parts. **Part I, “The Fundamentals of Machine Learning”**, covers the following topics:

- What machine learning is, what problems it tries to solve, and the main categories and fundamental concepts of its systems
- The steps in a typical machine learning project
- Learning by fitting a model to data
- Optimizing a cost function
- Handling, cleaning, and preparing data
- Selecting and engineering features
- Selecting a model and tuning hyperparameters using cross-validation
- The challenges of machine learning, in particular underfitting and overfitting (the bias/variance trade-off)

- The most common learning algorithms: linear and polynomial regression, logistic regression, k -nearest neighbors, support vector machines, decision trees, random forests, and ensemble methods
- Reducing the dimensionality of the training data to fight the “curse of dimensionality”
- Other unsupervised learning techniques, including clustering, density estimation, and anomaly detection

Part II, “Neural Networks and Deep Learning”, covers the following topics:

- What neural nets are and what they’re good for
- Building and training neural nets using TensorFlow and Keras
- The most important neural net architectures: feedforward neural nets for tabular data, convolutional nets for computer vision, recurrent nets and long short-term memory (LSTM) nets for sequence processing, encoder–decoders and transformers for natural language processing (and more!), autoencoders, generative adversarial networks (GANs), and diffusion models for generative learning
- Techniques for training deep neural nets
- How to build an agent (e.g., a bot in a game) that can learn good strategies through trial and error, using reinforcement learning
- Loading and preprocessing large amounts of data efficiently
- Training and deploying TensorFlow models at scale

The first part is based mostly on Scikit-Learn, while the second part uses TensorFlow and Keras.

CAUTION

Don’t jump into deep waters too hastily: while deep learning is no doubt one of the most exciting areas in machine learning, you should master the fundamentals first. Moreover, most problems can be solved quite well using simpler techniques such as random forests and ensemble methods (discussed in **Part I**). deep learning is best suited for complex problems such as image recognition, speech recognition, or natural language processing, and it requires a lot of data, computing power, and patience (unless you can leverage a pretrained neural network, as you will see).

Changes Between the First and the Second Edition

If you have already read the first edition, here are the main changes between the first and the second edition:

- All the code was migrated from TensorFlow 1.x to TensorFlow 2.x, and I replaced most of the low-level TensorFlow code (graphs, sessions, feature columns, estimators, and so on) with much simpler Keras code.
- The second edition introduced the Data API for loading and preprocessing large datasets, the distribution strategies API to train and deploy TF models at scale, TF Serving and Google Cloud AI Platform to productionize models, and (briefly) TF Transform, TFLite, TF Addons/Seq2Seq, TensorFlow.js, and TF Agents.
- It also introduced many additional ML topics, including a new chapter on unsupervised learning, computer vision techniques for object detection and semantic segmentation, handling sequences using convolutional neural networks (CNNs), natural language processing (NLP) using recurrent neural networks (RNNs), CNNs and transformers, GANs, and more.

See <https://homl.info/changes2> for more details.

Changes Between the Second and the Third Edition

If you read the second edition, here are the main changes between the second and the third edition:

- All the code was updated to the latest library versions. In particular, this third edition introduces many new additions to Scikit-Learn (e.g., feature name tracking, histogram-based gradient boosting, label propagation, and more). It also introduces the *Keras Tuner* library for hyperparameter tuning, Hugging Face's *Transformers* library for natural language processing, and Keras's new preprocessing and data augmentation layers.
- Several vision models were added (ResNeXt, DenseNet, MobileNet, CSPNet, and EfficientNet), as well as guidelines for choosing the right one.
- **Chapter 15** now analyzes the Chicago bus and rail ridership data instead of generated time series, and it introduces the ARMA model and its variants.
- **Chapter 16** on natural language processing now builds an English-to-Spanish translation model, first using an encoder–decoder RNN, then using a transformer model. The chapter also covers language models such as Switch Transformers, DistilBERT, T5, and PaLM (with chain-of-thought prompting). In addition, it introduces vision transformers (ViTs) and gives an overview of a few transformer-based visual models, such as data-efficient image transformers (DeiT), Perceiver, and DINO, as well as a brief overview of some large multimodal models, including CLIP, DALL·E, Flamingo, and GATO.
- **Chapter 17** on generative learning now introduces diffusion models, and shows how to implement a denoising diffusion probabilistic model (DDPM) from scratch.

- **Chapter 19** migrated from Google Cloud AI Platform to Google Vertex AI, and uses distributed Keras Tuner for large-scale hyperparameter search. The chapter now includes TensorFlow.js code that you can experiment with online. It also introduces additional distributed training techniques, including PipeDream and Pathways.
- In order to allow for all the new content, some sections were moved online, including installation instructions, kernel principal component analysis (PCA), mathematical details of Bayesian Gaussian mixtures, TF Agents, and former appendices A (exercise solutions), C (support vector machine math), and E (extra neural net architectures).

See <https://homl.info/changes3> for more details.

Other Resources

Many excellent resources are available to learn about machine learning. For example, Andrew Ng's **ML course on Coursera** is amazing, although it requires a significant time investment.

There are also many interesting websites about machine learning, including Scikit-Learn's exceptional **User Guide**. You may also enjoy **Dataquest**, which provides very nice interactive tutorials, and ML blogs such as those listed on **Quora**.

There are many other introductory books about machine learning. In particular:

- Joel Grus's *Data Science from Scratch*, 2nd edition (O'Reilly), presents the fundamentals of machine learning and implements some of the main algorithms in pure Python (from scratch, as the name suggests).
- Stephen Marsland's *Machine Learning: An Algorithmic Perspective*, 2nd edition (Chapman & Hall), is a great introduction to machine learning, covering a wide range of topics in depth with code examples in Python (also from scratch, but using NumPy).
- Sebastian Raschka's *Python Machine Learning*, 3rd edition (Packt Publishing), is also a great introduction to machine learning and leverages Python open source libraries (Pylearn 2 and Theano).
- François Chollet's *Deep Learning with Python*, 2nd edition (Manning), is a very practical book that covers a large range of topics in a clear and concise way, as you might expect from the author of the excellent Keras library. It favors code examples over mathematical theory.
- Andriy Burkov's *The Hundred-Page Machine Learning Book* (self-published) is very short but covers an impressive range of topics, introducing them in approachable terms without shying away from the math equations.

- Yaser S. Abu-Mostafa, Malik Magdon-Ismail, and Hsuan-Tien Lin's *Learning from Data* (AMLBook) is a rather theoretical approach to ML that provides deep insights, in particular on the bias/variance trade-off (see [Chapter 4](#)).
- Stuart Russell and Peter Norvig's *Artificial Intelligence: A Modern Approach*, 4th edition (Pearson), is a great (and huge) book covering an incredible amount of topics, including machine learning. It helps put ML into perspective.
- Jeremy Howard and Sylvain Gugger's *Deep Learning for Coders with fastai and PyTorch* (O'Reilly) provides a wonderfully clear and practical introduction to deep learning using the fastai and PyTorch libraries.

Finally, joining ML competition websites such as [Kaggle.com](#) will allow you to practice your skills on real-world problems, with help and insights from some of the best ML professionals out there.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

`Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values or by values determined by context.

Punctuation

To avoid any confusion, punctuation appears outside of quotes throughout the book. My apologies to the purists.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

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NOTE

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Never in my wildest dreams did I imagine that the first and second editions of this book would get such a large audience. I received so many messages from readers, many asking questions, some kindly pointing out errata, and most sending me encouraging words. I cannot express how grateful I am to all these readers for their tremendous support. Thank you all so very much! Please do not hesitate to **file issues on GitHub** if you find errors in the code examples (or just to ask questions), or to submit **errata** if you find errors in the text. Some readers also shared how this book helped them get their first job, or how it helped them solve a concrete problem they were working on. I find such feedback incredibly motivating. If you find this book helpful, I would love it if you could share your story with me, either privately (e.g., via **LinkedIn**) or publicly (e.g., tweet me at **@aureliengeron** or write an **Amazon review**).

Huge thanks as well to all the wonderful people who offered their time and expertise to review this third edition, correcting errors and making countless suggestions. This edition is so much better thanks to them: Olzhas Akpambetov, George Bonner, François Chollet, Siddha Ganju, Sam Goodman, Matt Harrison, Sasha Sobran, Lewis Tunstall, Leandro von Werra, and my dear brother Sylvain. You are all amazing!

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Last but not least, I am infinitely grateful to my beloved wife, Emmanuelle, and to our three wonderful children, Alexandre, Rémi, and Gabrielle, for encouraging me to work hard on this book. Their insatiable curiosity was priceless: explaining some of the most difficult concepts in this book to my wife and children helped me clarify my thoughts and directly improved many parts of it. Plus, they keep bringing me cookies and coffee, who could ask for more?

¹ Geoffrey E. Hinton et al., “A Fast Learning Algorithm for Deep Belief Nets”, *Neural Computation* 18 (2006): 1527–1554.

² Despite the fact that Yann LeCun’s deep convolutional neural networks had worked well for image recognition since the 1990s, although they were not as general-purpose.

Part I. The Fundamentals of Machine Learning

Chapter 1. The Machine Learning Landscape

Not so long ago, if you had picked up your phone and asked it the way home, it would have ignored you—and people would have questioned your sanity. But machine learning is no longer science fiction: billions of people use it every day. And the truth is it has actually been around for decades in some specialized applications, such as optical character recognition (OCR). The first ML application that

really became mainstream, improving the lives of hundreds of millions of people, took over the world back in the 1990s: the *spam filter*. It's not exactly a self-aware robot, but it does technically qualify as machine learning: it has actually learned so well that you seldom need to flag an email as spam anymore. It was followed by hundreds of ML applications that now quietly power hundreds of products and features that you use regularly: voice prompts, automatic translation, image search, product recommendations, and many more.

Where does machine learning start and where does it end? What exactly does it mean for a machine to *learn* something? If I download a copy of all Wikipedia articles, has my computer really learned something? Is it suddenly smarter? In this chapter I will start by clarifying what machine learning is and why you may want to use it.

Then, before we set out to explore the machine learning continent, we will take a look at the map and learn about the main regions and the most notable landmarks: supervised versus unsupervised learning and their variants, online versus batch learning, instance-based versus model-based learning. Then we will look at the workflow of a typical ML project, discuss the main challenges you may face, and cover how to evaluate and fine-tune a machine learning system.

This chapter introduces a lot of fundamental concepts (and jargon) that every data scientist should know by heart. It will be a high-level overview (it's the only chapter without much code), all rather simple, but my goal is to ensure everything is crystal clear to you before we continue on to the rest of the book. So grab a coffee and let's get started!

TIP

If you are already familiar with machine learning basics, you may want to skip directly to **Chapter 2**. If you are not sure, try to answer all the questions listed at the end of the chapter before moving on.

What Is Machine Learning?

Machine learning is the science (and art) of programming computers so they can *learn from data*.

Here is a slightly more general definition:

[Machine learning is the] field of study that gives computers the ability to learn without being explicitly programmed.

—Arthur Samuel, 1959

And a more engineering-oriented one:

A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

—Tom Mitchell, 1997

Your spam filter is a machine learning program that, given examples of spam emails (flagged by users) and examples of regular emails (nospam, also called “ham”), can learn to flag spam. The examples that the system uses to learn are called the *training set*. Each training example is called a *training instance* (or *sample*). The part of a machine learning system that learns and makes predictions is called a *model*. Neural networks and random forests are examples of models.

In this case, the task T is to flag spam for new emails, the experience E is the *training data*, and the performance measure P needs to be defined; for example, you can use the ratio of correctly classified emails. This particular performance measure is called *accuracy*, and it is often used in classification tasks.

If you just download a copy of all Wikipedia articles, your computer has a lot more data, but it is not suddenly better at any task. This is not machine learning.

Why Use Machine Learning?

Consider how you would write a spam filter using traditional programming techniques (Figure 1-1):

1. First you would examine what spam typically looks like. You might notice that some words or phrases (such as “4U”, “credit card”, “free”, and “amazing”) tend to come up a lot in the subject line. Perhaps you would also notice a few other patterns in the sender’s name, the email’s body, and other parts of the email.
2. You would write a detection algorithm for each of the patterns that you noticed, and your program would flag emails as spam if a number of these patterns were detected.
3. You would test your program and repeat steps 1 and 2 until it was good enough to launch.

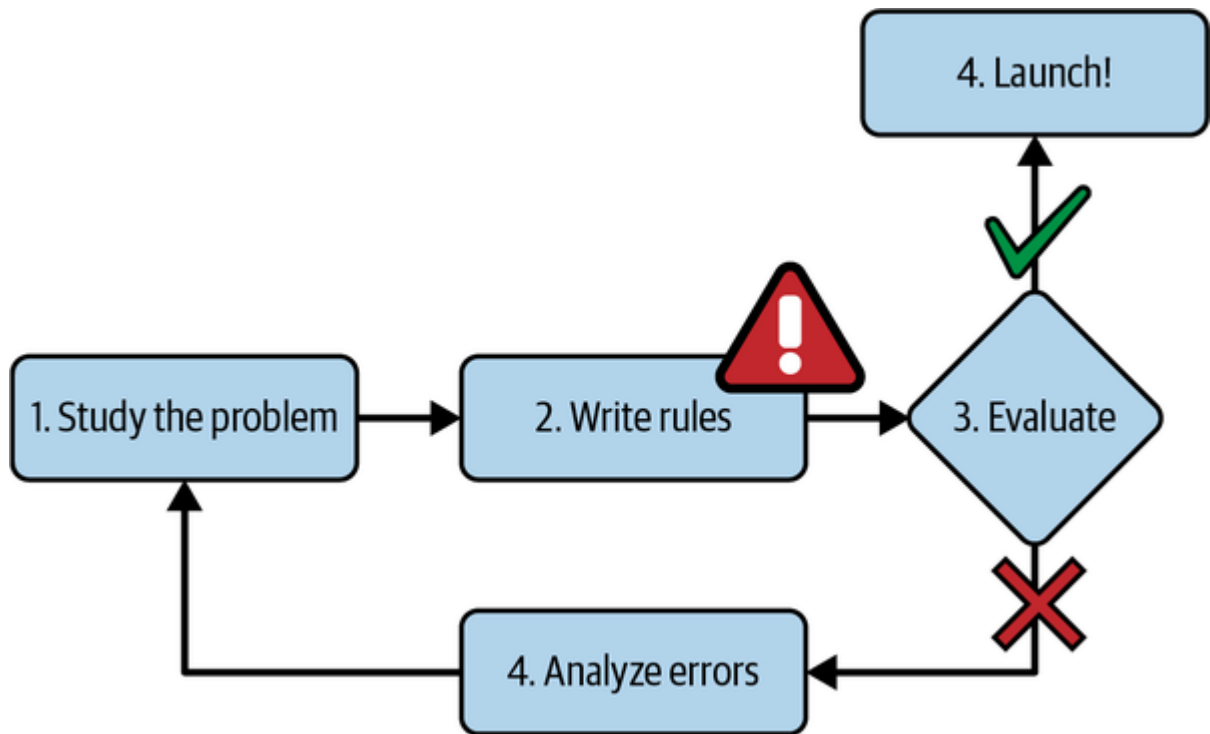


Figure 1-1. The traditional approach

Since the problem is difficult, your program will likely become a long list of complex rules—pretty hard to maintain.

In contrast, a spam filter based on machine learning techniques automatically learns which words and phrases are good predictors of spam by detecting unusually frequent patterns of words in the spam examples compared to the ham examples (Figure 1-2). The program is much shorter, easier to maintain, and most likely more accurate.

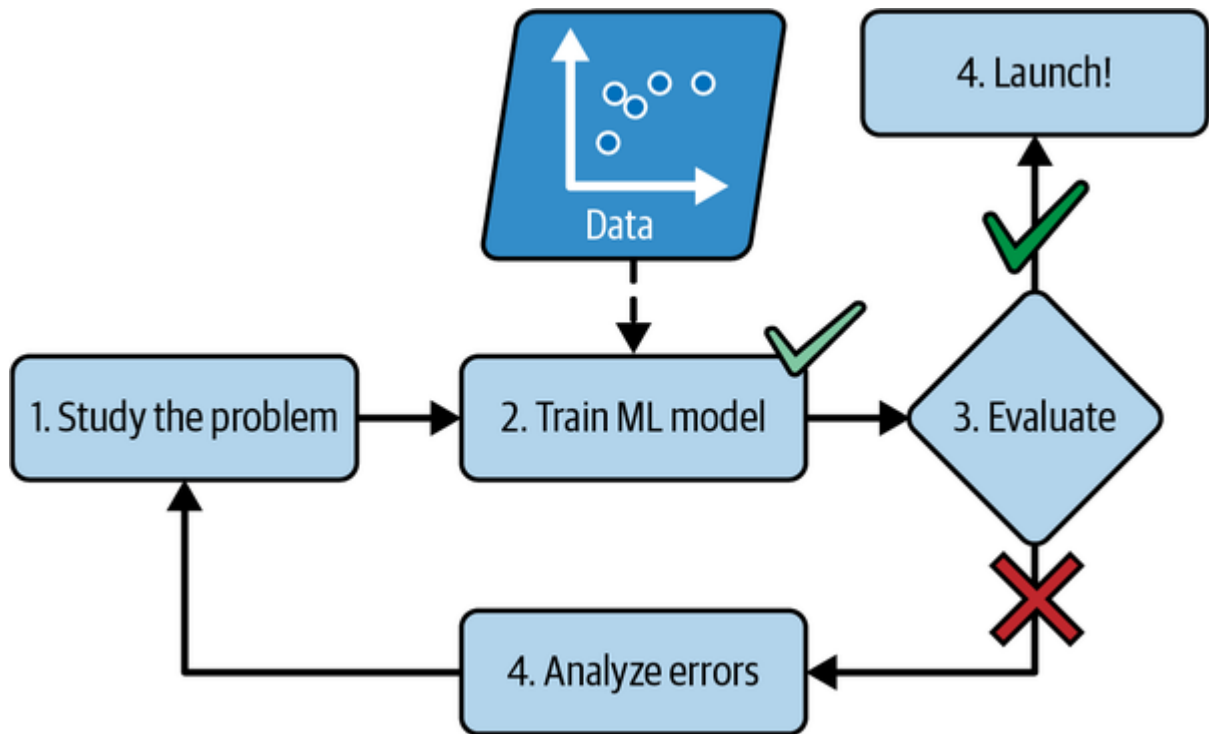


Figure 1-2. The machine learning approach

What if spammers notice that all their emails containing “4U” are blocked? They might start writing “For U” instead. A spam filter using traditional programming techniques would need to be updated to flag “For U” emails. If spammers keep working around your spam filter, you will need to keep writing new rules forever.

In contrast, a spam filter based on machine learning techniques automatically notices that “For U” has become unusually frequent in spam flagged by users, and it starts flagging them without your intervention (Figure 1-3).

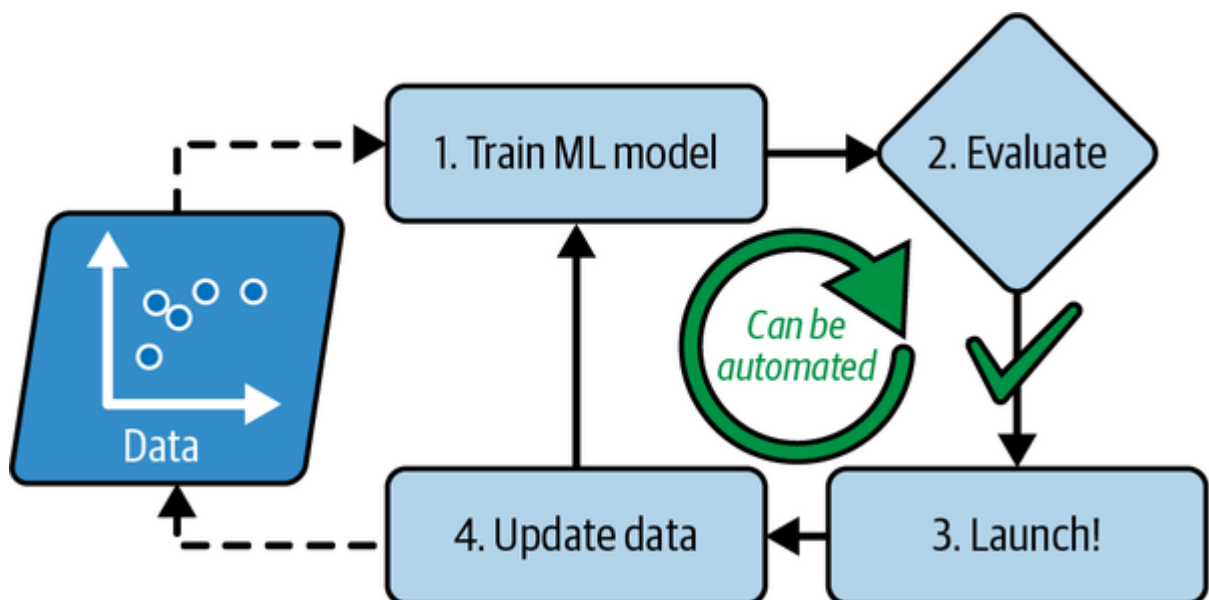


Figure 1-3. Automatically adapting to change

Another area where machine learning shines is for problems that either are too complex for traditional approaches or have no known algorithm. For example, consider speech recognition. Say you want to start simple and write a program capable of distinguishing the words “one” and “two”. You might notice that the word “two” starts with a high-pitch sound (“T”), so you could hardcode an algorithm that measures high-pitch sound intensity and use that to distinguish ones and twos—but obviously this technique will not scale to thousands of words spoken by millions of very different people in noisy environments and in dozens of languages. The best solution (at least today) is to write an algorithm that learns by itself, given many example recordings for each word.

Finally, machine learning can help humans learn (Figure 1-4). ML models can be inspected to see what they have learned (although for some models this can be tricky). For instance, once a spam filter has been trained on enough spam, it can easily be inspected to reveal the list of words and combinations of words that it believes are the best predictors of spam. Sometimes this will reveal unsuspected correlations or new trends, and thereby lead to a better understanding of the problem. Digging into large amounts of data to discover hidden patterns is called *data mining*, and machine learning excels at it.

This document was truncated here because it was created in the Evaluation Mode.

